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CERTIFICATE

This is to certify that the project report titled **"PREDICTION HEALTH INSURANCE COST"** submitted by **ROUSHAN KANT SINGH (25SCS1003002793) , KANHAIYA LAL YADAV (25SCS1003002780) , SUDHANSHU SINGH (25SCS1003002789) , MAYANK (25SCS1003002750)** in partial fulfilment of the requirements for the internship/project evaluation, is a Bonafide work carried out under my supervision.

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INTRODUCTION

This project focuses on predicting health insurance charges using machine learning techniques. The system allows users to input personal and medical attributes such as age, sex, BMI, number of children, smoking status, and region. Based on these factors, a trained regression model estimates the annual insurance cost.

To make the prediction process simple and interactive, a Streamlit-based web application is developed. Users can make single predictions, upload datasets for batch predictions, visualize data distributions, and understand model behavior using SHAP explanations. The system supports loading models dynamically, handling both default and uploaded model files.

Overall, this project provides an easy-to-use platform for understanding how different attributes influence health insurance costs and demonstrates practical implementation of machine learning in real-world decision-making.

PROBLEM STATEMENT

Health insurance costs vary widely depending on factors such as age, BMI, smoking habits, gender, region, and number of dependents. Individuals and insurance companies often find it difficult to estimate these charges accurately, leading to financial uncertainty and inefficient planning.

The problem is to develop a machine learning–based prediction system that can accurately estimate annual health insurance charges using user-provided inputs or a dataset. The system should preprocess data, apply a trained regression model, generate predictions, and present results in an interactive and user-friendly interface.

The goal is to create a tool that helps users understand how personal attributes impact insurance costs and assists stakeholders in making informed and data-driven decisions.

OBJECTIVES

1. To build a machine learning model capable of accurately predicting annual health insurance charges using key user features such as age, BMI, sex, children, smoking status, and region.

2. To preprocess and encode input data effectively so the model receives correctly formatted and meaningful features for prediction.

3. To develop an interactive Streamlit application that allows users to:

- Enter values for single prediction
- Upload a dataset for batch prediction
- Upload or use a saved model file

4. To provide data visualization features including distributions, correlations, and scatter plots to help users understand patterns in the dataset.

5. To implement model interpretability using SHAP values and feature importance so users can understand how different attributes influence predicted charges.

6. To enable exporting results, such as predicted charges for multiple entries, in a downloadable CSV format.

7. To create a user-friendly, efficient, and educational tool that demonstrates practical application of machine learning in real-world insurance cost prediction.

SYSTEM ARCHITECTURE

This project follows a simple and efficient architecture for predicting health insurance charges using a trained machine learning model. The system is built using Streamlit, scikit-learn, and pandas.

1. Overview

The system is designed to:

- Take user inputs (age, sex, BMI, children, smoker status, region)
- Preprocess and encode features
- Load a trained ML model (uploaded or local)
- Generate single or batch predictions
- Display visualizations and SHAP explanations

2. Architecture Components

a) User Interface (UI) - Streamlit

- Entire front-end runs in

  streamlit_insurance_app.py

- UI elements include:
  - Input form for single prediction
  - File uploader for CSV data
  - File uploader for a trained model
  - Visualizations (histograms, scatter plots, heatmaps)
  - SHAP model explanation section

b) Data Layer

- Loads dataset from:
  - Local file (insurance.csv)
  - OR uploaded CSV
- Uses pandas for cleaning, display, and batch predictions.

c) Model Layer

- Loads ML model from:

    - Uploaded .joblib / .pkl file
    - Local default model
    - Falls back to a dummy model if none provided
- Performs:
    - Preprocessing via prepare_features()
    - Running model.predict() for predictions

d) Visualization & Explainability

- Uses Matplotlib & Seaborn for charts
- Optional SHAP section for model interpretation
- Batch predictions can be exported as CSV

3. Data Flow

User → Streamlit UI

→ Uploads data/model or enters inputs

→ Feature preparation (one-hot encoding etc.)

→ Model prediction

→ Output: estimated charges + graphs + optional SHAP plots

4. Key Features

- Single prediction
- Batch prediction on uploaded dataset
- Auto-loading local dataset and model
- Visual insights (distribution, correlation, scatter plots)
- SHAP-based feature importance (optional)

5. Summary

Overall, the architecture is a simple, single-file Streamlit-based system where UI, data processing, and machine-learning inference all run together. It is ideal for academic projects, demos, and quick prototyping.

METHODOLOGY

The development of the Health Insurance Cost Prediction System follows a systematic methodology consisting of five major stages:

1. **Data Acquisition**

- The dataset (insurance.csv) is collected containing attributes such as age, sex, BMI, number of children, smoking status, region, and charges.
- The Streamlit app also allows users to upload their own CSV file for batch predictions.
- Data is loaded using pandas inside the application for further processing.

2. **Data Preprocessing**

- The dataset is checked for missing values, data types, and inconsistencies.
- Categorical variables (sex, smoker, region) are encoded using one-hot encoding.
- Numerical features such as age, BMI, and children are kept as continuous values.
- A dedicated function prepare_features() is used to transform user inputs or dataset values into the exact format required by the model.
- Preprocessing ensures that the input features match the model's expected structure.

3. **Model Design / Training**

- Machine learning regression algorithms such as Linear Regression and Random Forest Regressor are used for training.
- The model learns the relationship between user attributes and insurance charges.
- The trained model is saved as .joblib or .pkl file so it can be loaded into the Streamlit application.
- If no trained model is provided, a dummy fallback model is used for demonstration purposes.

4. **Application (UI) Development**

- A user-friendly Streamlit web application is developed to interact with the model.
- Key UI features include:
  - Single prediction form
  - Dataset upload for bulk prediction
  - Model upload option
  - Interactive charts (histograms, scatter plots, heatmaps)
  - Feature importance and SHAP explainability
- The app displays predictions instantly and allows users to download results as a CSV file.

5. **Testing & Evaluation**

- The prediction system is tested using single inputs and batch inputs.
- Plots and SHAP explanations help evaluate model behavior and understand feature impact.
- Any feature mismatch or model loading error is handled through validations and error messages.
- The system is checked for usability, accuracy, and performance across different inputs and datasets.

Technologies Used

1. Python – Core programming language used for building the prediction model and application logic.

2. Pandas – Used for data loading, cleaning, preprocessing, and handling CSV files.

3. NumPy – Supports numerical operations and array-based computations.

4. Scikit-learn – Used to train machine learning models such as Linear Regression and Random Forest.

5. Joblib / Pickle – Used for saving and loading trained machine learning models.

6. Streamlit – Framework used to build the interactive web application for predictions, data upload, and visualization.

7. Matplotlib & Seaborn – Used for creating plots such as histograms, scatter plots, and correlation heatmaps.

8. SHAP – Used to provide model explainability and visualize feature importance.

9. OS & Base64 Libraries – Used for file handling and exporting prediction results as downloadable CSV files.

Implementation & Code

The Health Insurance Cost Prediction System is implemented using Python, machine learning models, and a Streamlit web interface. The implementation consists of four main steps:

1. Data Loading

The dataset insurance.csv is loaded using pandas.

It contains the required features: age, sex, BMI, children, smoker, region, and charges.

```
df = pd.read_csv('insurance.csv')
```

2. Data Preprocessing

User inputs and dataset rows are converted into model-ready format using the prepare_features() function inside the Streamlit app.

Categorical features are one-hot encoded, and numerical features are kept as continuous values.

```
def prepare_features(age, sex, bmi, children, smoker, region):
    df_x = pd.DataFrame({
        'age':[age], 'sex':[sex], 'bmi':[bmi],
        'children':[children], 'smoker':[smoker], 'region':[region]
    })
    df_x['sex_male'] = (df_x['sex']=='male').astype(int)
    df_x['smoker_yes'] = (df_x['smoker']=='yes').astype(int)
    # region one-hot encoding
    ...
```

```
return df_x  # returns model-ready features
```

3. Model Loading & Prediction

The Streamlit app loads a trained machine learning model (.joblib / .pkl).

If no model is uploaded, a dummy model is used for demonstration.

```
model = joblib.load('model.joblib')   # or uploaded model
```

```
pred = model.predict(X)[0]
```

The prediction is displayed in the sidebar when the user clicks Predict.

4. Streamlit Application

- **The main interface (streamlit_insurance_app.py) provides:**
- **Single prediction form**
- **CSV upload for batch predictions**
- **Model upload**
- **Visualizations (histogram, scatter, heatmap)**
- **SHAP-based explanation (optional)**
- **CSV download for predicted results**

The application is started using:

```
streamlit run streamlit_insurance_app.py
```

Conclusion

The Health Insurance Cost Prediction System successfully demonstrates how machine learning can be applied to estimate insurance charges based on key personal and medical factors. By using regression models and an interactive Streamlit interface, the system provides users with quick, accurate, and easy-to-understand predictions.

The application also includes data visualization and SHAP-based model explainability, helping users understand how different features—such as age, BMI, and smoking habits—affect the final insurance cost. With support for both single and batch predictions, the system is flexible, user-friendly, and suitable for real-world educational and analytical use.

Overall, this project highlights the effective integration of machine learning, data preprocessing, and user interface design to create a practical and informative prediction tool.

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4. NumPy Documentation – Scientific Computing with Python

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5. Matplotlib Documentation – Visualization with Python

https://matplotlib.org/

6. Seaborn Documentation – Statistical Data Visualization

https://seaborn.pydata.org/

7. SHAP Documentation – Explainable AI – SHAP Values

https://shap.readthedocs.io/

8. Dataset: Medical Cost Personal Dataset (Insurance Dataset)

Commonly used for regression tasks and health insurance prediction studies.

9. Code structure and implementation inspired by the uploaded project files

– streamlit_insurance_app.py

– tempCodeRunnerFile.py (run command)